

## # Final Optional Refactor QA Report

### Field Value

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Branch branch-v1-final-optional-refactor-qa

Base commit 8f90105

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Scope Remaining optional installer refactor, exit-path hardening, typed error contracts

Score 100/100

Pass rate 4/4 automated gates

### ## Result Matrix

#### Gate Command Result Evidence

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Static analysis dart analyze PASS evidence/dart\_analyze.txt

Full repository tests dart test --concurrency=1 --reporter=expanded PASS, 318 tests  
evidence/dart\_test.txt

Targeted refactor regression tests installer service, argument, registry, path, and  
resolution tests PASS, 49 tests evidence/targeted\_refactor\_tests.txt

CLI integration tests dart test test/cli\_integration\_test.dart --concurrency=1  
--reporter=expanded PASS, 38 tests evidence/cli\_integration\_test.txt

### ## Issues Resolved

#### Issue Before After

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Remaining installer part surface Pubspec mutation, dry-run planning, and platform  
instruction writes lived inside installer part extensions. Added InstallerPubspecService,  
InstallerDryRunService, and InstallerPlatformService; part files now delegate to these  
services.

Pubspec mutation coupling Installer owned dependency, asset, and font writes directly.  
Pubspec edits are handled by InstallerPubspecService while preserving comments and  
formatting through PubspecEditor.

Dry-run logic coupling Dry-run plan construction and rendering were embedded in an installer  
part. InstallerDryRunService owns dependency-aware plan construction and dry-run output  
rendering.

Platform instruction coupling Platform marker writes and post-install note reporting were  
embedded beside alias generation. InstallerPlatformService owns platform targets, marker  
idempotency, and post-install note output. Alias generation remains isolated in the alias  
part.

Generic component resolution errors AddResolutionService threw generic Exception values and  
tests matched message text. Added ComponentResolutionException with message and token, and  
updated tests to assert the typed contract.

Path guard error typing Filesystem root escape and missing project root errors used generic  
exceptions. Added PathEscapeException and ProjectRootNotFoundException; root containment now  
uses canonical path boundary checks instead of prefix matching.

Library/service process exits Registry selection, file-kind parsing, version upgrade, and

studio service paths could call `exit()` directly. These paths now throw typed exceptions or return command-level exit codes; upgrade preserves the same numeric exit behavior through `runUpgradeCommand`.

Explicit local registry fallback A bad explicit `--registry-path` could fall through to auto-discovery. Explicit bad local paths now fail with `RegistryBootstrapException` and `ExitCodes.registryNotFound`.

## ## OpenCode Agent Work

Three OpenCode agents were run in read-only mode:

### Agent Focus Applied Result

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Refactor audit Remaining installer part extraction candidates Implemented the low-risk dry-run and platform extraction recommendations; also extracted pubspec mutation with direct tests.

Hygiene audit Generic exceptions, `exit()` usage, logger/print risks Fixed production-relevant exit paths and typed core/component/path errors. Broader logger-print normalization remains cleanup, not a functional blocker.

QA/report audit Final report and changelog structure Used the proposed evidence matrix and before/after inventory for this report and `changelog_report.md`.

## ## Residual Risk

No production-blocking CLI issue remains in this branch based on the verified gates. Remaining cleanup opportunities are architectural only:

- Further split high-coupling installer parts such as theme, remove, locale, and manifest logic after path resolution is extracted into a shared resolver abstraction.
- Standardize all application-layer `print()` output behind `CliLogger` or a dedicated output abstraction.
- Continue replacing generic exceptions in theme/schema/registry loaders with typed domain exceptions over future cleanup branches.

## ## Evidence Summary

- dart analyze: No issues found!
- Full suite: All tests passed! with 318 tests.
- Targeted refactor suite: All tests passed! with 49 tests.
- CLI integration suite: All tests passed! with 38 tests.